

TEMP-1 Temperature Sensor

Probe-ready temperature and humidity monitoring for Home Assistant

The Apollo TEMP-1 is a compact temperature and humidity sensor built for ESPHome and Home Assistant over Wi-Fi. An onboard AHT20-F measures ambient temperature and humidity, while a 3.5 mm jack accepts optional waterproof or food-safe probes for accurate readings exactly where you need them. It is controlled entirely on your local network with no cloud connection or subscription required.

TEMP-1 is offered in a USB-C model for continuous operation and a battery model (rechargeable RC-CR123A or 16340 cell, not included) for cable-free placement in locations without constant power. Onboard RGB and buzzer feedback provide real-time alerts, and all firmware and CAD files are open source.

Key Specifications

Product	TEMP-1 Temperature Sensor
Manufacturer	Apollo Automation (Versailles, KY, USA)
Onboard sensor	AHT20-F — temperature & humidity
Probe interface	3.5 mm jack — 1 optional external probe (sold separately)
Microcontroller	ESP32
Firmware	ESPHome — open source
Connectivity	Wi-Fi (2.4 GHz) + Bluetooth
Power	USB-C (Non-Battery) or RC-CR123A / 16340 rechargeable (Battery model; cell not included)
Battery life	Up to ~6 months with optimized sleep settings
Onboard feedback	RGB LED + piezo buzzer
Local control	Yes — no cloud or subscription required
In the box	Assembled sensor with case; Apollo sticker

Onboard Components

Component	Function
AHT20-F <i>Aosong</i>	Onboard ambient temperature and humidity sensor
3.5 mm jack	Accepts one optional external temperature probe (sold separately)
RGB LED + buzzer	Customizable visual and audible real-time alerts
ESP32 + Bluetooth	Wi-Fi and Bluetooth connectivity for local smart-home integration

Temperature Probe Options

Probe (sold separately)	Sensor	Max temp	Notes
1.5 m (~5 ft) flat cable	DS18B20	85 °C / 185 °F	Waterproof, submersible
20 cm (~8 in) flat cable	DS18B20	85 °C / 185 °F	Waterproof, submersible
1 m (~3 ft) stainless steel	NTC	204 °C / 400 °F	Food-safe; not dishwasher safe

Two probes are waterproof flat-cable submersible types whose thin cable can pass through fridge or freezer door seals; one is a food-safe stainless-steel probe. All probes are sold separately. The TEMP-1 supports multiple probes via configuration — see the Multiple Probe Guide.

Configurations & Pricing

Configuration	Power	Price (USD)
Non-Battery	USB-C, continuous	\$24.99
Battery (battery not included)	RC-CR123A / 16340 rechargeable	\$25.99

Recommended battery sources: imrbatteries.com (USA), nkon.nl (EU). Battery cell is not included with the Battery model.

Features & Use Cases

- **Food safety:** monitor fridge/freezer temps (flat probe fits through door seals) and track cooking with the food-safe probe.
- **Aquatic:** monitor aquarium, pool, or hot-tub water temperature with the waterproof probe.
- **Home & building:** automate attic fans on temperature thresholds and monitor ambient temps to optimize HVAC.
- **Outdoor & remote:** use the battery model in greenhouses, sheds, barns, or storage where power isn't available.
- **Safety alerts:** trigger the buzzer or Home Assistant alerts on overheat in server rooms or electronics enclosures.

Important Notes

- **Use a probe for best accuracy:** the onboard AHT20-F can be affected by heat from the ESP chip; enabling sleep settings reduces this for accurate onboard temperature and humidity readings.
- **Probes sold separately:** the 3.5 mm jack accepts the optional probes listed above; none are included in the box.
- **Food probe care:** the stainless-steel food-safe probe is rated to 204 °C / 400 °F and is not dishwasher safe.
- **Battery not included:** the Battery model requires a user-supplied rechargeable RC-CR123A or 16340 cell.

Documentation & Resources

- **Product page:** apolloautomation.com/products/temp-1
- **Wiki / setup guide:** wiki.apolloautomation.com/products/temp1
- **Multiple probe guide:** wiki.apolloautomation.com/products/temp1/addons/multiple-probes
- **Source & CAD files:** github.com/ApolloAutomation/TEMP-1
- **Printable mounts:** [Printables — Apollo TEMP-1](#)
- **Community:** [Apollo Automation Discord](#)